

CURSORS

Scenario 1: GenerateMonthlyStatements

```
SET SERVEROUTPUT ON;
```

```
DECLARE
```

```
    CURSOR GenerateMonthlyStatements IS
```

```
        SELECT c.CustomerID,
```

```
               c.Name,
```

```
               t.TransactionID,
```

```
               t.TransactionDate,
```

```
               t.Amount,
```

```
               t.TransactionType
```

```
        FROM Customers c
```

```
        JOIN Accounts a
```

```
            ON c.CustomerID = a.CustomerID
```

```
        JOIN Transactions t
```

```
            ON a.AccountID = t.AccountID
```

```
        WHERE EXTRACT(MONTH FROM t.TransactionDate) = EXTRACT(MONTH  
FROM SYSDATE)
```

```
        AND EXTRACT(YEAR FROM t.TransactionDate) = EXTRACT(YEAR FROM  
SYSDATE);
```

```
    v_CustomerID Customers.CustomerID%TYPE;
```

```
    v_Name Customers.Name%TYPE;
```

```
    v_TransactionID Transactions.TransactionID%TYPE;
```

```
    v_Date Transactions.TransactionDate%TYPE;
```

```
    v_Amount Transactions.Amount%TYPE;
```

```
    v_Type Transactions.TransactionType%TYPE;
```

```
BEGIN
```

```
    OPEN GenerateMonthlyStatements;
```

```
    LOOP
```

```
        FETCH GenerateMonthlyStatements
```

```
        INTO v_CustomerID, v_Name, v_TransactionID, v_Date, v_Amount, v_Type;
```

```
        EXIT WHEN GenerateMonthlyStatements%NOTFOUND;
```

```

        DBMS_OUTPUT.PUT_LINE(
            'Customer: ' || v_Name ||
            ', Transaction ID: ' || v_TransactionID ||
            ', Date: ' || TO_CHAR(v_Date,'DD-MON-YYYY') ||
            ', Amount: ' || v_Amount ||
            ', Type: ' || v_Type
        );
    END LOOP;

    CLOSE GenerateMonthlyStatements;
END;
/

```

The screenshot displays the FreeSQL online SQL editor. The left sidebar shows a 'Navigator' with 'My Schema' and a list of tables: ACCOUNTS, AUDITLOG, CUSTOMERS, EMPLOYEES, LOANS, and TRANSACTIONS. The main editor area shows a PL/SQL script with the following code:

```

1  SET SERVEROUTPUT ON;
2
3  DECLARE
4      CURSOR GenerateMonthlyStatements IS
5          SELECT c.CustomerID,
6                 c.Name,
7                 t.TransactionID,
8                 t.TransactionDate,
9                 t.Amount,
10                t.TransactionType
11          FROM Customers c
12          JOIN Accounts a
13              ON c.CustomerID = a.CustomerID
14          JOIN Transactions t
15              ON a.AccountID = t.AccountID
16          WHERE EXTRACT(MONTH FROM t.TransactionDate) = EXTRACT(MONTH FROM SYSDATE)
17                AND EXTRACT(YEAR FROM t.TransactionDate) = EXTRACT(YEAR FROM SYSDATE);

```

Below the script, the 'Script output' tab is active, showing the results of the query:

```

Customer: John Doe, Transaction ID: 1001, Date: 07-JUL-2026, Amount: 500, Type: Deposit
Customer: John Doe, Transaction ID: 301, Date: 17-JUL-2026, Amount: 500, Type: Deposit
Customer: John Doe, Transaction ID: 302, Date: 17-JUL-2026, Amount: 1000, Type: Deposit
Customer: Jane Smith, Transaction ID: 1002, Date: 10-JUL-2026, Amount: 300, Type: Withdrawal
Customer: Emily Davis, Transaction ID: 1004, Date: 11-JUL-2026, Amount: 200, Type: Deposit
Customer: Michael Lee, Transaction ID: 1005, Date: 04-JUL-2026, Amount: 400, Type: Withdrawal

```

At the bottom, the status indicates: 'PL/SQL procedure successfully completed.' and 'Elapsed: 00:00:00.024'.

Scenario 2: ApplyAnnualFee

```
SET SERVEROUTPUT ON;
```

```
DECLARE
    CURSOR ApplyAnnualFee IS
        SELECT AccountID
        FROM Accounts;

    v_AccountID Accounts.AccountID%TYPE;

BEGIN
    OPEN ApplyAnnualFee;

    LOOP
        FETCH ApplyAnnualFee
        INTO v_AccountID;

        EXIT WHEN ApplyAnnualFee%NOTFOUND;

        UPDATE Accounts
        SET Balance = Balance - 200
        WHERE AccountID = v_AccountID;

        DBMS_OUTPUT.PUT_LINE(
            'Annual fee deducted from Account ' || v_AccountID
        );

    END LOOP;

    CLOSE ApplyAnnualFee;

    COMMIT;
END;
/
```

Navigator Files

My Schema

Tables

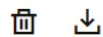
Search

- ACCOUNTS
- AUDITLOG
- CUSTOMERS
- EMPLOYEES
- LOANS
- TRANSACTIONS

[SQL Worksheet]*      Aa 

```
1 SET SERVEROUTPUT ON;
2
3 DECLARE
4     CURSOR ApplyAnnualFee IS
5         SELECT AccountID
6             FROM Accounts;
7
8     v_AccountID Accounts.AccountID%TYPE;
9
10 BEGIN
11     OPEN ApplyAnnualFee;
12
13     LOOP
14         FETCH ApplyAnnualFee
15             INTO v_AccountID;
16
17         EXIT WHEN ApplyAnnualFee%NOTFOUND;
```

Query result Script output DBMS output Explain Plan SQL history



Annual fee deducted from Account 101
Annual fee deducted from Account 102
Annual fee deducted from Account 103
Annual fee deducted from Account 104
Annual fee deducted from Account 105

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.009

Query result			Script output	DBMS output	Explain Plan	SQL history
			Download ▾ Execution time: 0.004 seconds			
	ACCOUNTID	BALANCE				
1	101	305				
2	102	14300				
3	103	22020				
4	104	4850				
5	105	12800				

Scenario 3: ApplyLoanInterestRates

SET SERVEROUTPUT ON;

DECLARE

CURSOR UpdateLoanInterestRates IS
 SELECT LoanID
 FROM Loans;

v_LoanID Loans.LoanID%TYPE;

BEGIN

OPEN UpdateLoanInterestRates;

LOOP

FETCH UpdateLoanInterestRates
 INTO v_LoanID;

EXIT WHEN UpdateLoanInterestRates%NOTFOUND;

UPDATE Loans
 SET InterestRate = InterestRate - 0.5
 WHERE LoanID = v_LoanID;

```

DBMS_OUTPUT.PUT_LINE(
    'Interest updated for Loan ID ' || v_LoanID
);

END LOOP;

CLOSE UpdateLoanInterestRates;

COMMIT;
END;
/

```

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Navigator Files

My Schema

Tables

Search

ACCOUNTS

AUDITLOG

CUSTOMERS

EMPLOYEES

LOANS

TRANSACTIONS

[SQL Worksheet]*

```

10 BEGIN
27 END LOOP;
28
29 CLOSE UpdateLoanInterestRates;
30
31 COMMIT;
32 END;
33 /

```

Query result Script output DBMS output Explain Plan SQL history

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.009

SQL> DECLARE
CURSOR UpdateLoanInterestRates IS
SELECT LoanID
FROM Loans;...

Show more...

Interest updated for Loan ID 201
Interest updated for Loan ID 202
Interest updated for Loan ID 203
Interest updated for Loan ID 204
Interest updated for Loan ID 205

PL/SQL procedure successfully completed.

Query result

Script output

DBMS output

Explain Plan

SQL history



Download



Execution time: 0.004 seconds

	LOANID	INTERESTRATE
1	201	4.5
2	202	5.5
3	203	5.5
4	204	5
5	205	5